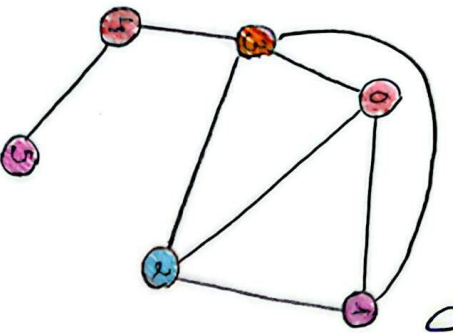


Vertex Coloring with minimum number of colors color 1. txt



$k=1 \Rightarrow [0]$ available color $[-1, -1, -1, -1, -1, -1]$ ^{color}
 $\exists T \mid \begin{array}{l} v=0 \quad c=0 \quad \text{valid color (uncolored neighbours)} \\ v=1 \quad c=0 \quad \text{invalid, no more colors (0 colored)} \end{array}$ $[0, -1, -1, -1, -1, -1]$ ^{color}

$\Rightarrow \text{False}$

$k=2 \Rightarrow [0, 1]$ available colors $[-1, -1, -1, -1, -1, -1]$ ^{color}

$\exists T \mid \begin{array}{l} v=0 \quad c=0 \quad \text{valid color (uncolored neighbours)} \\ v=1 \quad c=0 \quad \text{invalid (0 is colored with } c=0) \end{array}$ $[0, -1, -1, -1, -1, -1]$ ^{color}

$\exists T \mid \begin{array}{l} v=1 \quad c=0 \quad \text{invalid (0 is colored with } c=0) \\ v=2 \quad c=1 \quad \text{valid color } [0, 1, -1, -1, -1, -1] \end{array}$ ^{color}

$\exists T \mid \begin{array}{l} v=2 \quad c=0 \quad \text{invalid (0 colored)} \\ v=3 \quad c=1 \quad \text{invalid, no more colors} \end{array}$

$\Rightarrow \text{False (1 colored)}$

$\Rightarrow \text{False (1 colored)}$

$k=3 \Rightarrow [0, 1, 2]$ available colors $[-1, -1, -1, -1, -1, -1]$ ^{color}

$v=0 \quad c=0 \quad \text{valid color } [0, -1, -1, -1, -1, -1]$

$v=1 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{valid color } [0, 1, -1, -1, -1, -1]$

$v=2 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{invalid (1 colored)}$

$c=2 \quad \text{valid color } [0, 1, 2, -1, -1, -1]$

$v=3 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{invalid (1 colored)}$

$c=2 \quad \text{invalid, no more colors (2 colored)}$

$\Rightarrow \text{False}$

$k=4 \Rightarrow [0, 1, 2, 3]$ available colors $[-1, -1, -1, -1, -1, -1]$ ^{color}

$v=0 \quad c=0 \quad \text{valid color } [0, -1, -1, -1, -1, -1]$

$v=1 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{valid } [0, 1, -1, -1, -1, -1]$

$v=2 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{invalid (1 colored)}$

$c=2 \quad \text{valid } [0, 1, 2, -1, -1, -1]$

$v=3 \quad c=0 \quad \text{invalid (0 colored)}$

$c=1 \quad \text{invalid (1 colored)}$

$c=2 \quad \text{invalid (2 colored)}$

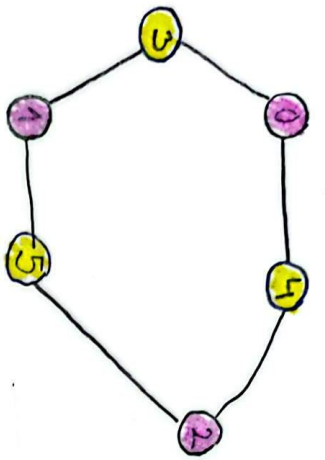
$c=3 \quad \text{valid } [0, 1, 2, 3, -1, -1]$

$v=4 \quad c=0 \quad \text{valid (no neighb. w color=0)} [0, 1, 2, 3, 0, -1]$

$v=5 \quad c=0 \quad \text{invalid (4 colored)}$

$c=1 \quad \text{valid (no neighb. w color=1)} [0, 1, 2, 3, 0, 1]$

$\Rightarrow \text{True}$



$k=1$. $[0]$ available color $[-1, -1, -1, -1, -1, -1]$

BT $\left| \begin{array}{ll} v=0 & c=0 \text{ valid } [0, -1, -1, -1, -1, -1] \\ v=1 & c=0 \text{ invalid (0 colored)} \end{array} \right.$
 no more colors

$\Rightarrow$ false

$k=2$ $[0, 1]$ available colors $[-1, -1, -1, -1, -1, -1]$

$v=0$ $c=0$ valid $[0, -1, -1, -1, -1, -1]$

$v=1$ $c=0$ valid $[0, 0, -1, -1, -1, -1]$

$v=2$ $c=0$ valid $[0, 0, 0, -1, -1, -1]$

$v=3$ $c=0$ invalid (0 colored)

$c=1$ valid $[0, 0, 0, 1, -1, -1]$

$v=4$ $c=0$ invalid (0, 2 colored)

$c=1$ valid $[0, 0, 0, 1, 1, -1]$

$v=5$ $c=0$ invalid (1, 2 colored)

$c=1$ valid $[0, 0, 0, 1, 1, 1]$

$\Rightarrow$ True